

Qatar-2b

R-0676

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_260328 · created 2026-07-27T19:04:03+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461127.8387 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0 +/- 0.032
Transit depth (Rp/Rs)^2	6.7e-06 +/- 0.0017 [%]
Orbital Inclination (inc)	85.45 +/- 1.71
Airmass coefficient 1 (a1)	0.603 +/- 0.031
Airmass coefficient 2 (a2)	0.372 +/- 0.045
Scatter in the residuals of the lightcurve fit is	5.15 %
Best Comparison Star	#2 - [469, 373]
Optimal Aperture	8.84
Optimal Annulus	21.35
Transit Duration (day)	0.0568 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.0 $\pm$ 0.032 vs 0.16327 (deviation 5.1 σ)
Duration fit vs published	0.0568 d vs 0.0767 d (deviation 2.21 σ)
Mid-time O-C	-0.1 min
Depth SNR	0.0
Residual scatter	5.15 %

Light curve

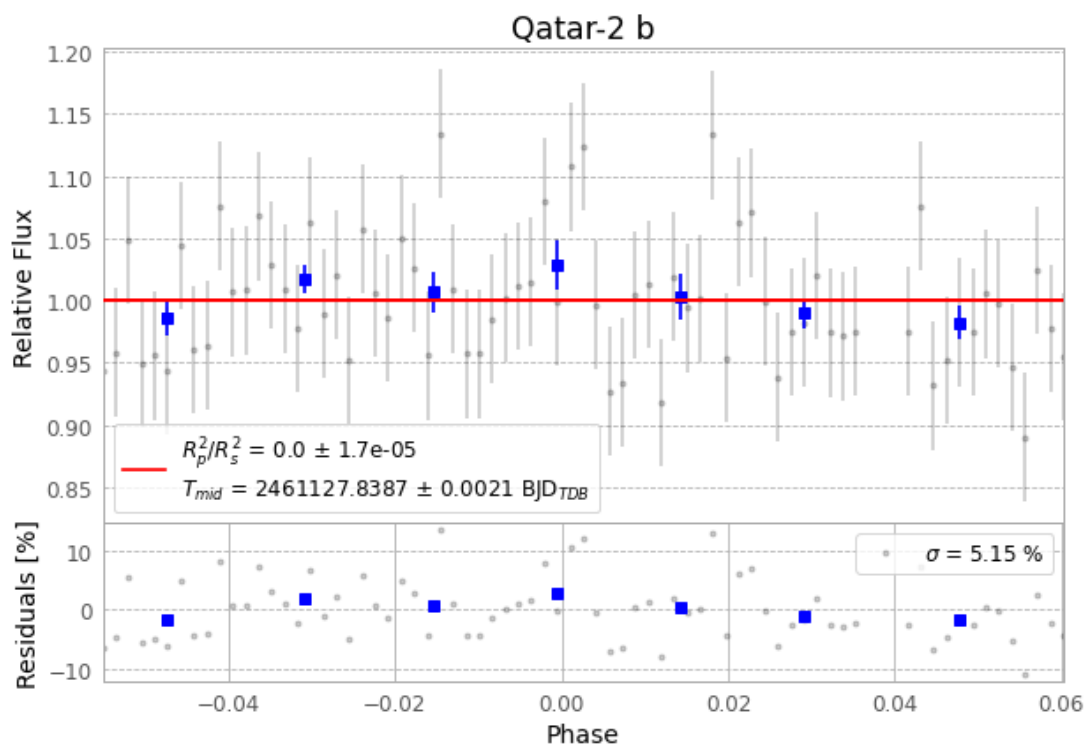


Figure 1 — FinalLightCurve_Qatar-2 b_2026-03-27.png (SHA-256 9ebe093fd3487106...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [435, 358], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 90b335016c4cb0f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

75 FITS frames (49 MB), Qatar-2260328061216.FITS → Qatar-2260328095416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-260328.log (eb52c780a48b...), FinalLightCurve_Qatar-2 b_2026-03-27.png (9ebe093fd348...), FinalLightCurve_Qatar-2 b_2026-03-27.csv (be633ab71edb...)

Compute resources

Wall time 195.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 19:04Z.